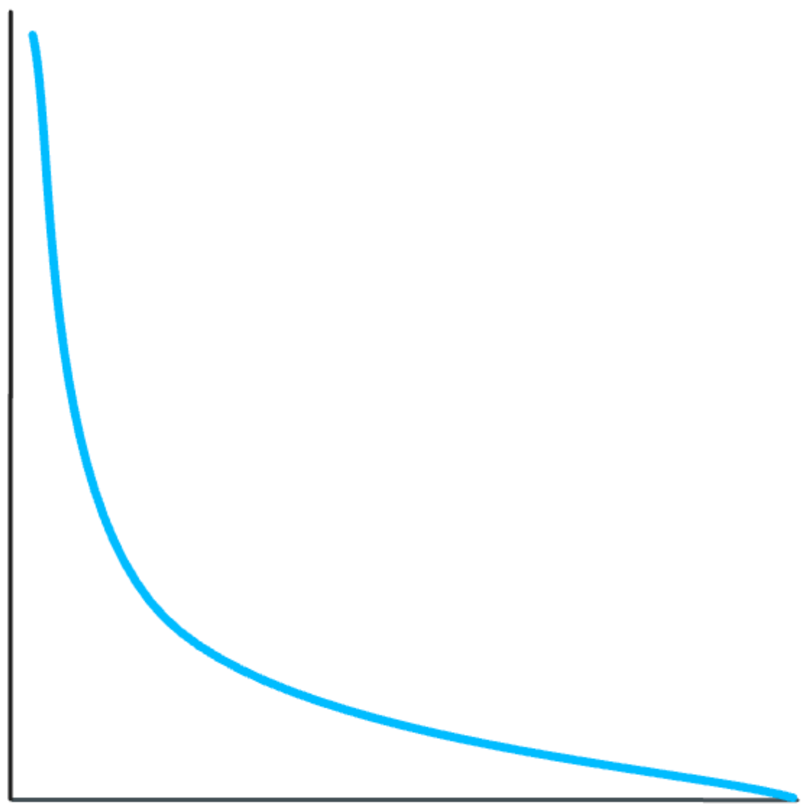
Log Loss

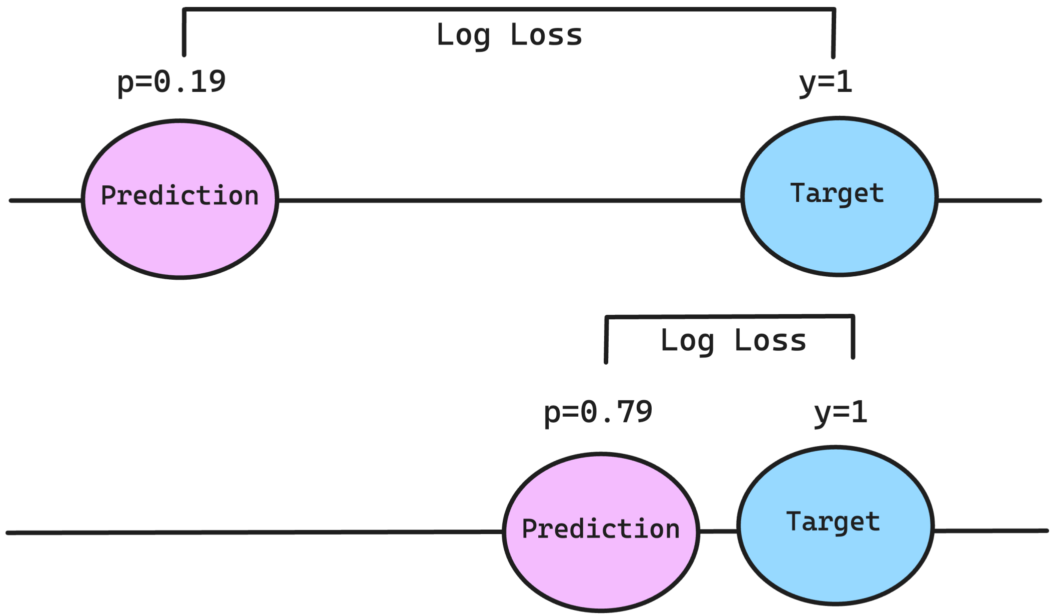


Log Loss

Log loss, also known as logistic loss or cross-entropy loss, is a common metric used in model training and evaluation of machine learning models working with binary outcomes[1] (e.g., credit default vs. no credit default)

Log Loss measures the performance of a predictive model whose output is a probability value between 0 and 1. Log loss penalizes false predictions (being too “far off” from the ground truth) by assigning a higher Log Loss score ➡️

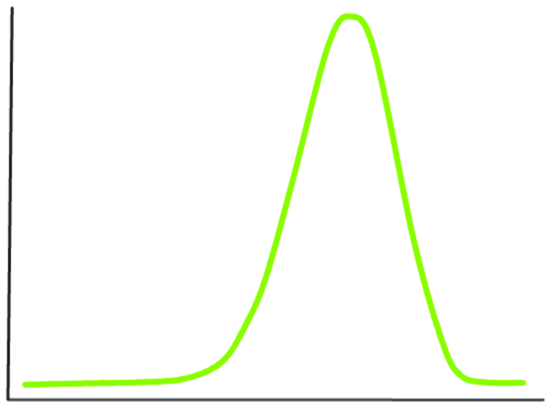
When working with this metric, we consider **lower Log Loss** indicating **better performance**[2], but there's no universal threshold for what constitutes a "good" log loss value unlike metrics such as ROC-AUC or the Gini score

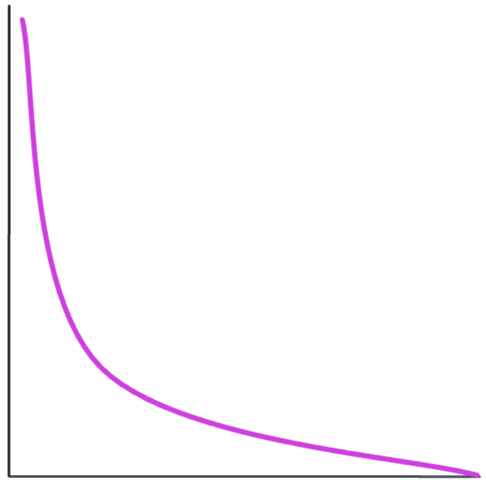


1. https://dasha.ai/en-us/blog/log-loss-function
2. https://analyticsindiamag.com/all-you-need-to-know-about-log-loss-in-machine-learning/

Contents

- Likelihood
- Binomial Likelihood
- Binomial Log-Likelihood
- Bernoulli Likelihood & Log-Likelihood
- From Binomial to Bernoulli Likelihood
- Log-Likelihood & MLE
- GBDT Example: Log-Likelihood
- Cumulative Log-Likelihood
- Training Loop for 30 GBDT Iterations
- From Log-Likelihood to Log Loss
- Log Loss
- Log Loss & Cross-Entropy
- Appendix (UBC Zoology: 06. Likelihood)





Likelihood

Imagine we have a dataset with 32 experiments (trials) where in 23 cases, wasps 🐝 chose the mated female butterfly 🦋

We would like to know how likely is it that this is not due to a random choice?[1]

Pieris Brassicae

1. Example from here: https://www.zoology.ubc.ca/biol548/lecturepdf/06.Likelihood.pdf

Binomial Likelihood

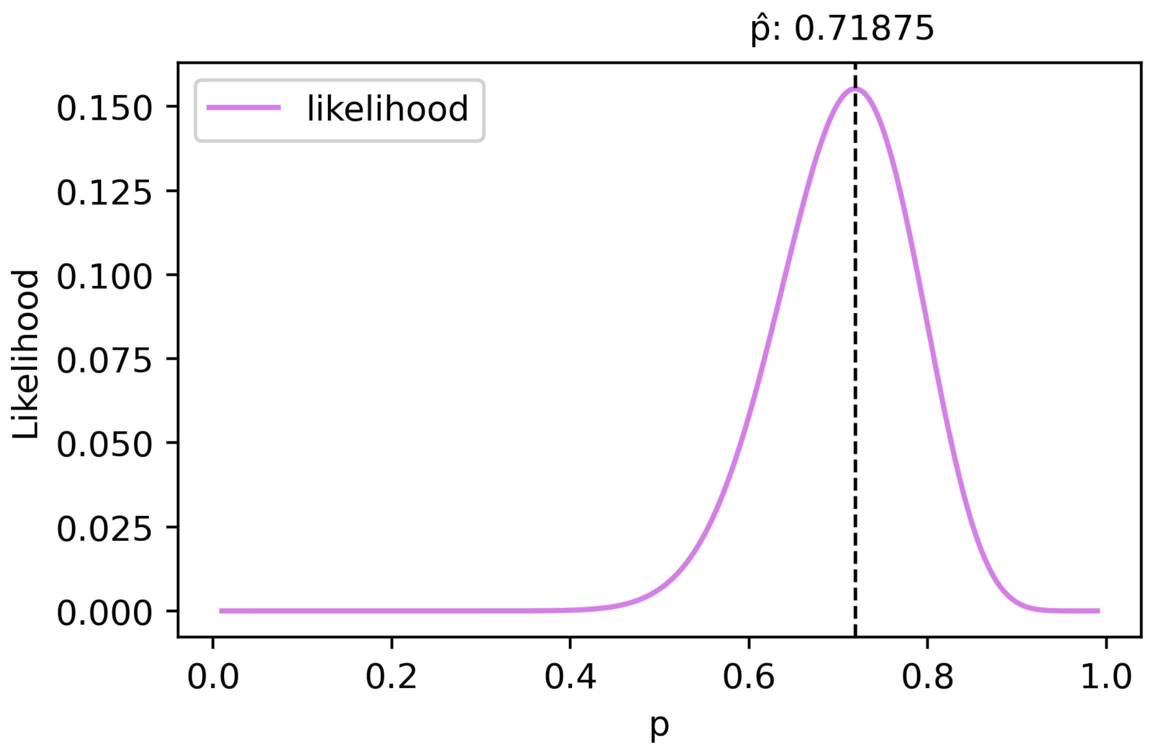
Binomial likelihood is an important starting point in understanding Log Loss based on this example

The formula below shows a likelihood function where:

- $L(p)$ = likelihood
- n = total number of trials
- k = the number of successes
- (n choose k) = a binomial coefficient
- p = the estimate (probability)

$$L(p) = \binom{n}{k} p^k (1-p)^{n-k}$$

The binomial likelihood is used to determine **the most probable value** of p. For our data, 32 trials and 23 successes, maximum likelihood is at p = 23/32 = 71.8%[1]



1. Example from here: https://www.zoology.ubc.ca/biol548/lecturepdf/06.Likelihood.pdf

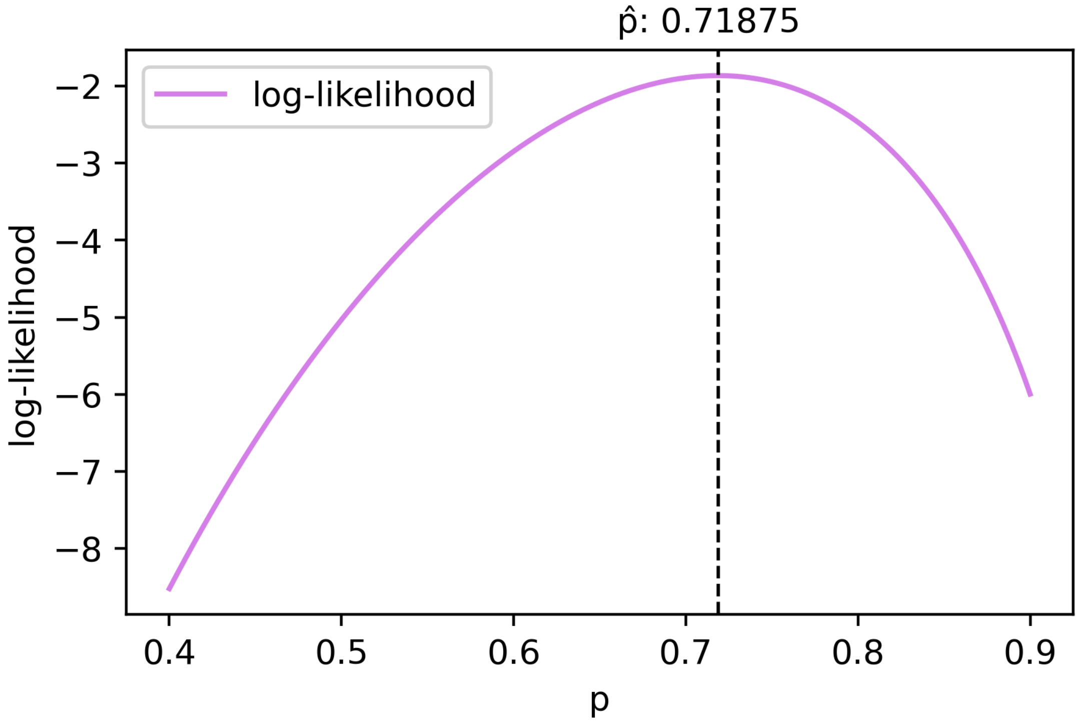
Binomial Log-Likelihood

If we use a log form, the binomial likelihood formula becomes:

- log $L(p)$ = log-likelihood
- n = total number of trials
- k = the number of successes
- (n choose k) = a binomial coefficient
- p = the estimate (probability)

$$\log L(p) = \log \binom{n}{k} + k\log(p) + (n-k)\log(1-p)$$

Similarly we can iterate over values of p and find **the most probable value** that resembles our data, in this case the log likelihood is maximum at p = 71.8%[1]



1. Example from here: https://www.zoology.ubc.ca/biol548/lecturepdf/06.Likelihood.pdf

Bernoulli Likelihood & Log-Likelihood

Bernoulli likelihood is a special case of a binomial likelihood where the number of trials, n, is fixed at 1. Bernoulli likelihood can be understood as a version of binomial likelihood on observation level

The binomial coefficient is no longer part of the likelihood function (1) and similarly for Bernoulli log-likelihood (2)

$$L(p) = \prod_{i=1}^{n} p^{\sum_{i=1}^{n} x_i}(1-p)^{n-\sum_{i=1}^{n} x_i} \quad (1)$$

$$LL(p) = \sum_{i=1}^{n} x_i \log(p) + (n - \sum_{i=1}^{n} x_i) \log(1-p) \quad (2)$$

We can compare the formulas for Binomial and Bernoulli likelihoods in the next slide ➡️

From Binomial to Bernoulli Likelihood

The Bernoulli likelihood is a binomial likelihood case with one observation per trial. This allows us to reframe our initial problem as a Bernoulli scenario by representing data at the level of individual observations

Consider a scenario involving a die roll: out of 12 trials 🎲, we observe 3 successes (which we define as getting a 5 ⚄)

Binomial likelihood	
Successes (⚄)	3
Trials (🎲)	12
Combination	220
p	**0.250**
p^k	0.01563
(1-p)^n-k	0.07508
Binomial L	**0.258**
Binomial LL	**-1.354**
Bernoulli LL	**-6.748**

Difference
is due to the log
of combination

Bernoulli likelihood				
Trial	**Outcome**	**p**	**1 - p**	**LL**
1	1	25.00%	75.00%	-1.386
2	0	25.00%	75.00%	-0.288
3	0	25.00%	75.00%	-0.288
4	0	25.00%	75.00%	-0.288
5	0	25.00%	75.00%	-0.288
6	0	25.00%	75.00%	-0.288
7	0	25.00%	75.00%	-0.288
8	1	25.00%	75.00%	-1.386
9	0	25.00%	75.00%	-0.288
10	1	25.00%	75.00%	-1.386
11	0	25.00%	75.00%	-0.288
12	0	25.00%	75.00%	-0.288
Σ12				**Σ-6.748**

From Binomial to Bernoulli Likelihood

We can confirm the log-likelihood calculations from logistic regression models using statsmodels in Python 🐍[1]

Binomial Log-likelihood

Model:	GLM	AIC:	4.7088
Link Function:	Logit	BIC:	-0.0000
Dependent Variable:	['successes', 'failures']	Log-Likelihood:	$\boxed{-1.3544}$
Date:	2024-02-21 21:55	LL-Null:	-1.3544
No. Observations:	1	Deviance:	-4.2598e-29
Df Model:	0	Pearson chi2:	4.44e-29
Df Residuals:	0	Scale:	1.0000
Method:	IRLS		

	Coef.	Std.Err.	z	P>|z|	[0.025	0.975]
Intercept	-1.0340	0.6275	-1.6479	0.0994	-2.2638	0.1958
X	-0.2585	0.1569	-1.6479	0.0994	-0.5659	0.0489

Bernoulli Log-likelihood

Generalized Linear Model Regression Results

Dep. Variable:	Success	**No. Observations:**	12
Model:	GLM	**Df Residuals:**	11
Model Family:	Binomial	**Df Model:**	0
Link Function:	Logit	**Scale:**	1.0000
Method:	IRLS	**Log-Likelihood:**	$\boxed{-6.7480}$
Date:	Wed, 21 Feb 2024	**Deviance:**	13.496
Time:	21:55:35	**Pearson chi2:**	12.0
No. Iterations:	4	**Pseudo R-squ. (CS):**	0.000
Covariance Type:	nonrobust		

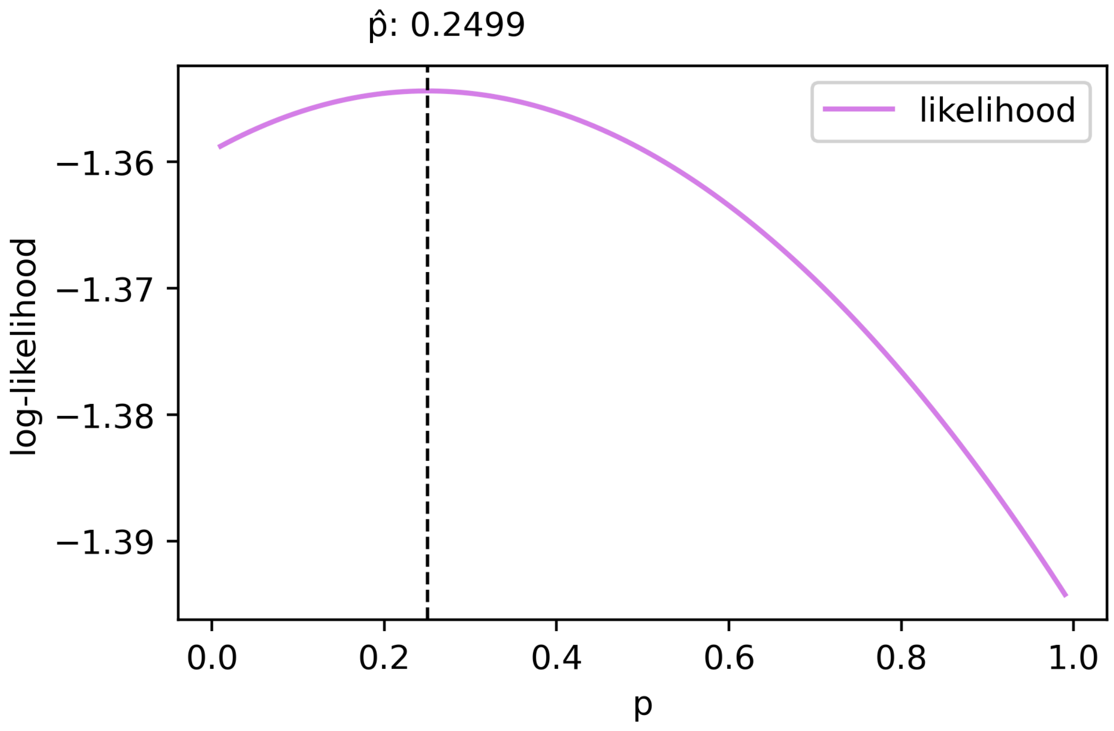
	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-1.0340	0.627	-1.648	0.099	-2.264	0.196
X	-0.2585	0.157	-1.648	0.099	-0.566	0.049

1. Logistic Regression: Bernoulli vs. Binomial Response Variables. https://stats.stackexchange.com/questions/144121/logistic-regression-bernoulli-vs-binomial-response-variables

From Binomial to Bernoulli Likelihood

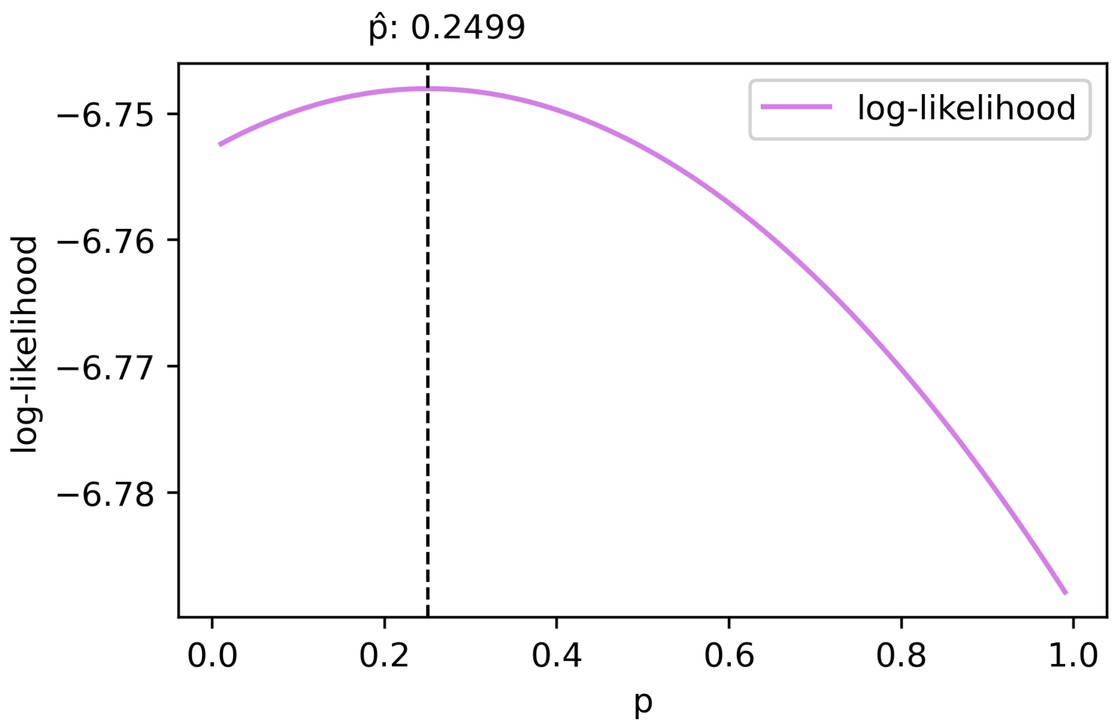
Using the intercept (a) and coefficient (b) from the two models we can visualize log-likelihood functions ⬇️

Binomial Log-likelihood



Maximum likelihood estimates: a = -1.034, b = -0.2585

Bernoulli Log-likelihood



Maximum likelihood estimates: a = -1.034, b = -0.2585

Remember that we don't have a log of the binomial coefficient for Bernoulli log-likelihood

Log-Likelihood & MLE

Log-likelihood is a way to measure how fit a model is for a dataset

When the parameters are estimated using the log-likelihood for the **Maximum Likelihood Estimation (MLE)**, each data point is added to the total log-likelihood

As the data can be viewed as an evidence that support the estimated parameters in MLE, this process can be interpreted as "support from independent evidence adds", and the log-likelihood is the "weight of evidence"[1]

Likelihood

$$L = \prod_{i=1}^{N} p_i^{y_i} (1 - p_i)^{1 - y_i}$$

Log-likelihood

$$LL = \sum_{i=1}^{N} y_i \log(p_i) + (1 - y_i) \log(1 - p_i)$$

Data point level

$$P(y) = \begin{cases} 1 - p_i & \text{for } y_i = 0 \\ p_i & \text{for } y_i = 1 \end{cases}$$

1. Likelihood function: https://en.wikipedia.org/wiki/Likelihood_function/

GBDT Example: Log-Likelihood

We can look at a gradient boosted decision tree (GBDT) model performing Logistic Regression to understand how log-likelihood can be calculated. We will take the first boosting iteration and look at the splits made by the model ⬇️

Feature	Sign	Split	Count	Events	NonEvents	EventRate	XAddEvidence
account_never_delinq_percent	<	98.0	2810.0	674.0	2136.0	0.239858	0.282129
account_never_delinq_percent	>=	98.0	4190.0	26.0	4164.0	0.006205	-0.635539

Table above shows that GBDT chose a particular feature (delinquency) and chose a threshold of 98 to create two distinct splits. The first split has a (credit default) event rate of 24% and the second one has 0.62%.
We can assume this is our model and we will use **the event rate (p)**[1] to calculate Log-likelihood (rounded):

$$LL_{split\ 1} = \Sigma\ 674 \times \log(0.239) + 2136 \times \log(1 - 0.239) = -1548$$

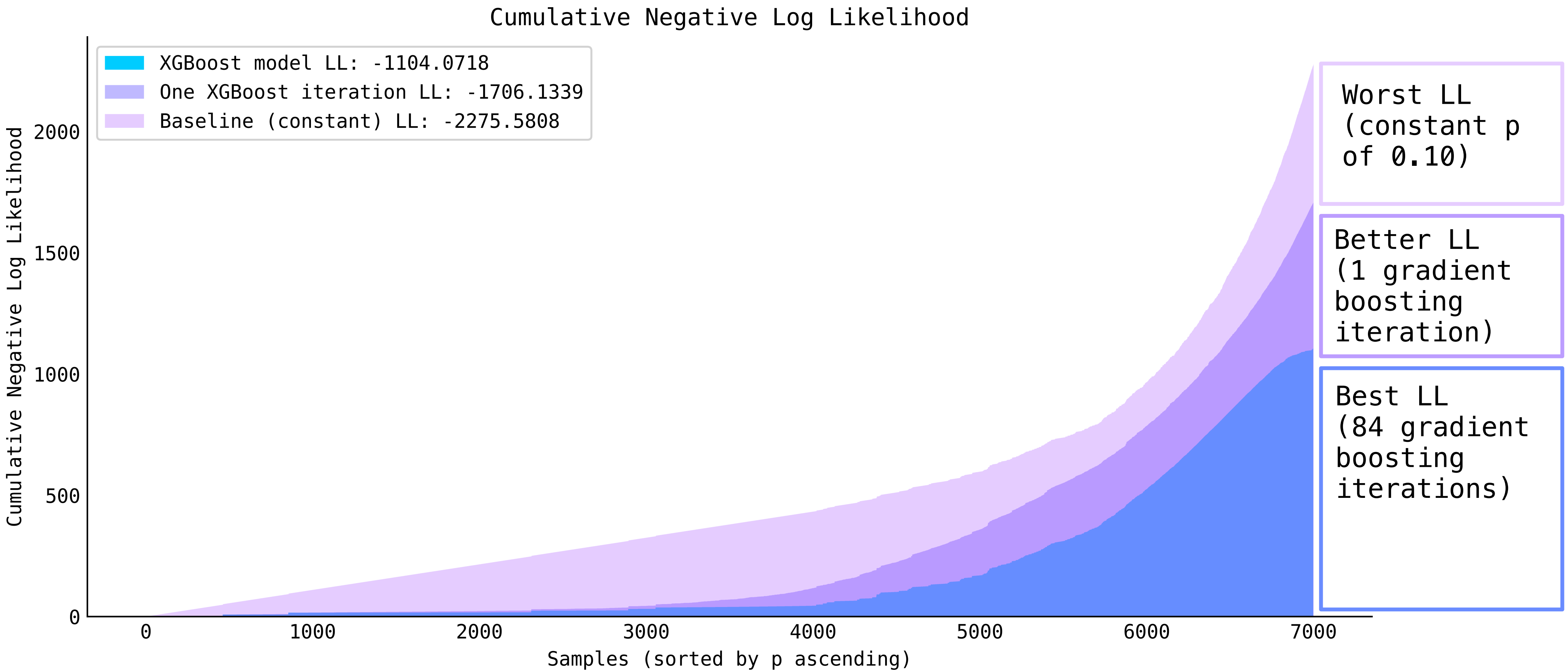
$$LL_{split\ 2} = \Sigma\ 26 \times \log(0.006) + 4164 \times \log(1 - 0.006) = -158$$

$$LL = LL_{split\ 1} + LL_{split\ 2} = (-1548) + (-158) = -1706$$

1. Technically it's not the prediction that GBDT models like XGBoost make in their first iteration, but we use this heuristic for simplification. You can read more about how XGBoost makes predictions in this tutorial: https://cengiz.me/posts/extreme-gradient-boosting/.

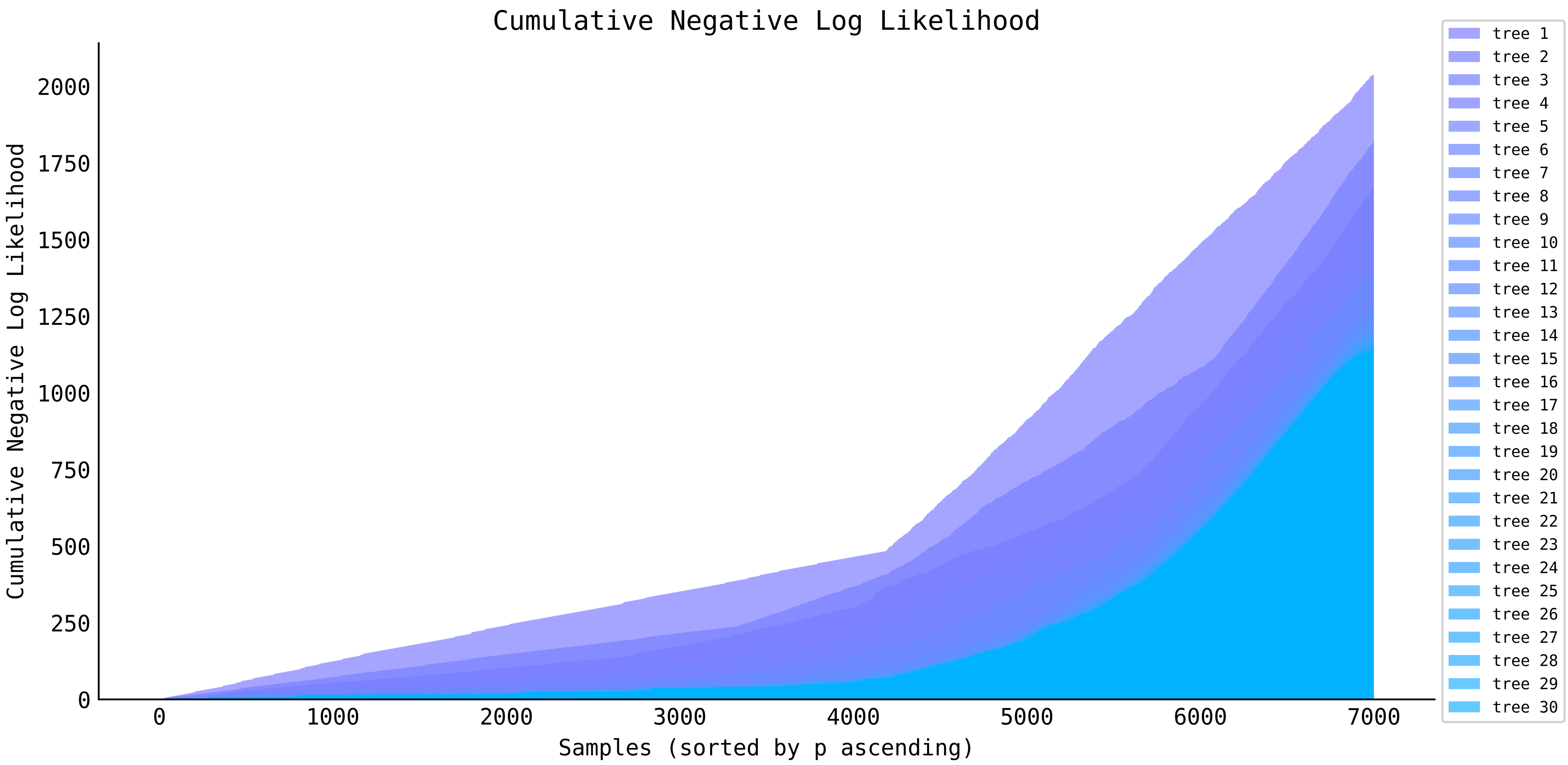
Cumulative Log-Likelihood

The value of Log-likelihood is not informative without a baseline and total count. Below we plot the Cumulative Log-likelihood from different models: GBDT model, first GBDT boosting iteration, and a constant prediction of p (0.10)

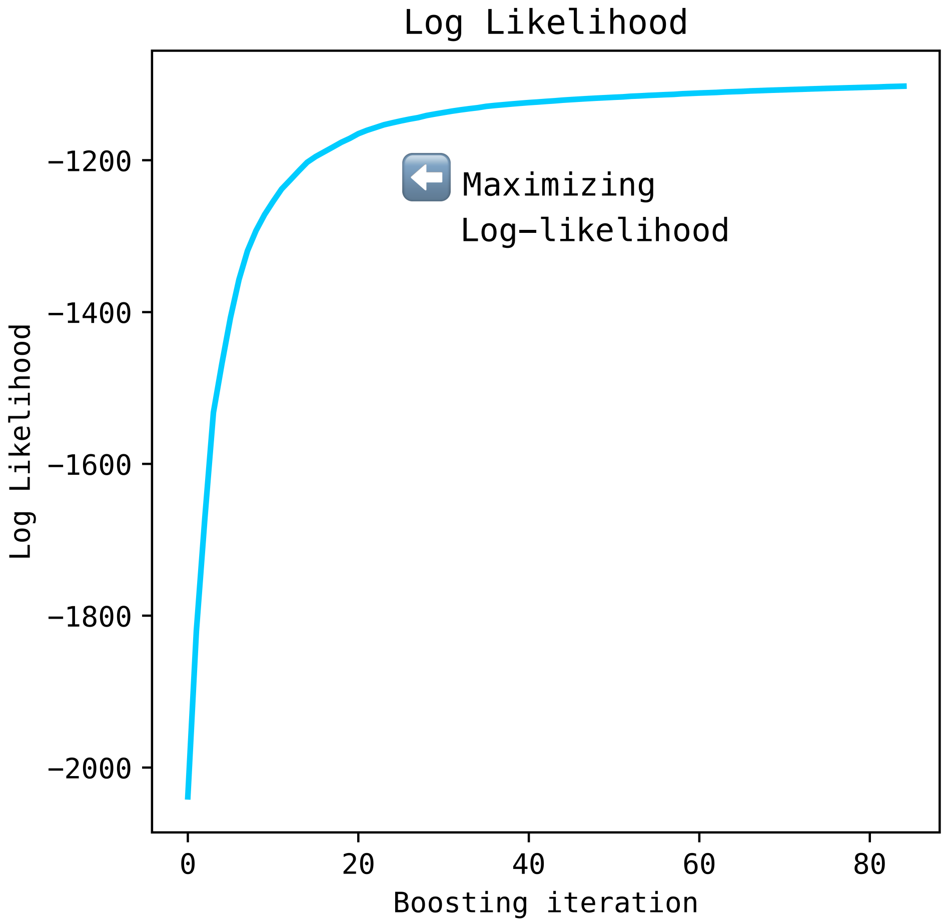


Training Loop for 30 GBDT Iterations

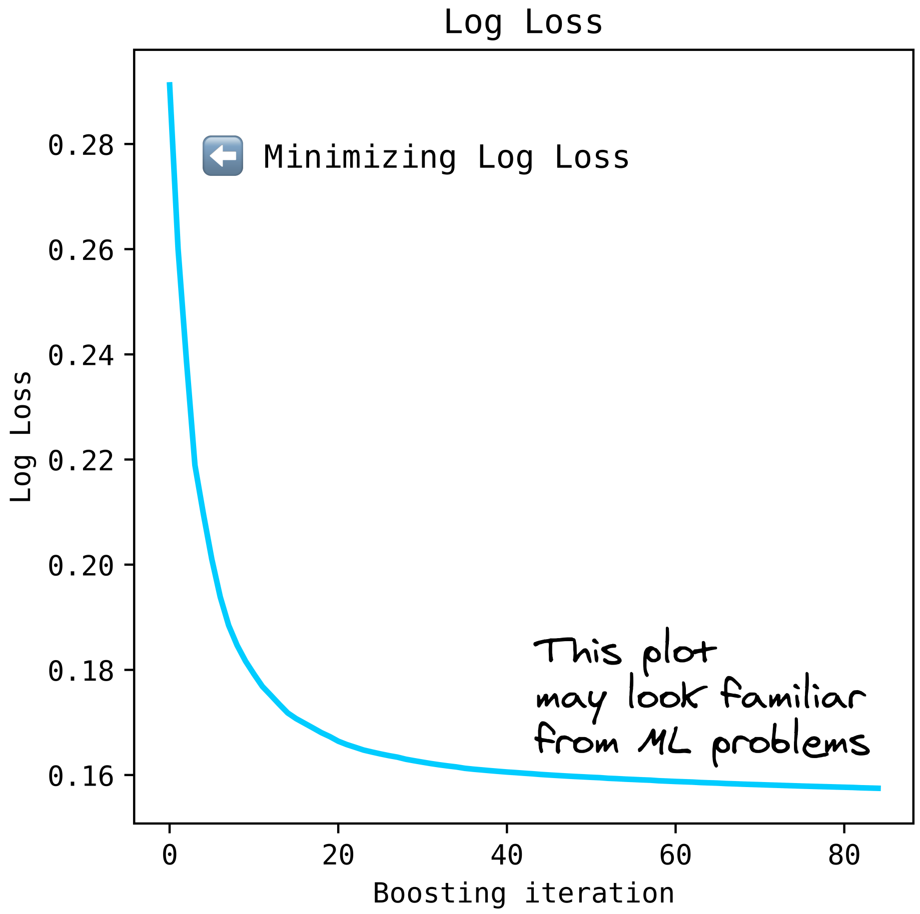
Notice how the cumulative negative Log-likelihood decreases with each gradient boosting iteration ⬇️



From Log-Likelihood to Log Loss



means



Log Loss

Log Loss is an average value of negative Log-likelihood (LL)[1] 💡

Since LL is negative, we need to add a minus sign to turn it into a positive value. In the example below, we can divide -(-1706.13) by 7000 or follow the conventional formula[1], -(1/7000 * -1706.13), to get a Log Loss of 24.37%

$$\text{Log Loss} = -\frac{1}{N}\sum_{i=1}^{N}\underbrace{[y_i \log(p_i) + (1-y_i)\log(1-p_i)]}_{\text{Log-likelihood (LL)}}$$

Feature	Sign	Split	Count	NonEvents	Events	EventRate	LL
account_never_delinq_percent	<	98	2810	2136	674	23.99%	-1548.07
account_never_delinq_percent	>=	98	4190	4164	26	0.62%	-158.06

LL	-1706.13
Total count	7000
Log Loss	**24.37%**

1. SciPy API reference: https://docs.scipy.org/doc/scipy/reference/generated/scipy.special.xlogy.html

Log Loss & Cross-Entropy

Another interesting question is why is Log Loss also called Cross-Entropy?

Shannon entropy is a concept introduced by Claude Shannon. Entropy quantifies the uncertainty of information[1]. To get Log Loss from entropies we can weight entropies by the absolute counts and divide it by the total counts

$$Entropy_{events} = -\, p_i \times \log(p_i)$$

$$Entropy_{non\text{-}events} = -\, (1 - p_i) \times \log(1 - p_i)$$

$$Entropy_i = Entropy_{events} + Entropy_{non\text{-}events}$$

Feature	Sign	Split	Count	NonEvents	Events	EventRate	Entropy NonEvents	Entropy Events	Entropy	Entropy × count
account_never_delinq_percent	<	98	2810	2136	674	23.99%	0.21	0.34	0.55	1548.1
account_never_delinq_percent	>=	98	4190	4164	26	0.62%	0.01	0.03	0.04	158.1

Total count	7000
Sum Entropy × count	1706.1
Log Loss	**24.37%**

1. Entropy (Information Theory): https://en.wikipedia.org/wiki/Entropy_(information_theory)

Link to the Code

Appendix

https://www.zoology.ubc.ca/biol548/lecturepdf/06.Likelihood.pdf

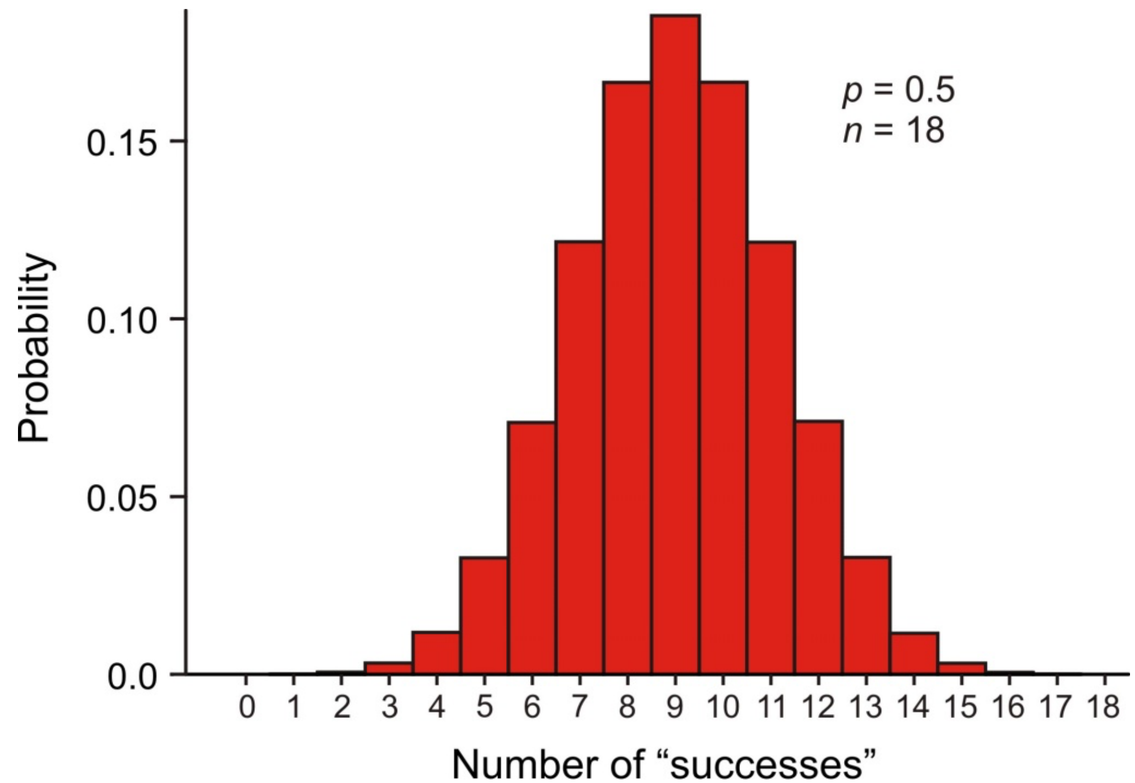
Example: binomial distribution

The *binomial distribution* is the probability distribution of the number of "successes" in *n* independent trials, when the probability of success *p* is the same in each trial.

$$\Pr[Y \text{ successes}] = \binom{n}{Y} p^{Y} (1-p)^{n-Y}$$

$\binom{n}{Y}$ counts up the different ways of getting *Y* successes and *n* – *Y* failures (e.g., S-S-F; S-F-S; F-S-S)

Graph shows Pr[0], Pr[1] , Pr[2], …
when *p* = 0.50 and *n* = 18



What is conditional probability

The *conditional probability* of an event is the probability of that event occurring given that a condition is met. "|" symbol used to indicate "given"

The probability that the second child born to a couple is a girl, given that their first child was a girl,

Pr[second child is girl | first child is girl]

Other conditional probabilities:

Pr[we see an elephant today | we are in the Serengeti]

Pr[we see an elephant today | we are in Manhattan]

Pr[12 successes in 18 trials | p = 0.50]

Pr[12 successes in 18 trials | p = 0.10]

What is likelihood

Likelihood is a conditional probability.

The likelihood of a population parameter equaling a specific value, given the data, is the probability of obtaining the observed data *given* that the population parameter equals the specific value.

L[parameter = ρ | data] = Pr[data | parameter = ρ]

L[p = 0.10 | 12 successes out of 18 trials] = Pr[12 successes in 18 trials | p = 0.10]

Law of Likelihood:

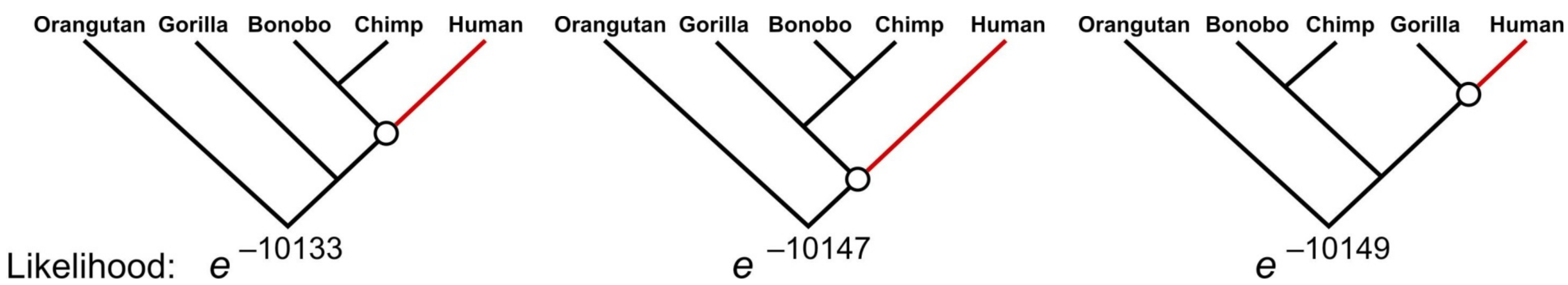
The extent to which data supports one parameter value or hypothesis against another is equal to the ratio of their likelihoods (difference in their log-likelihoods)

Method invented by R. A. Fisher when a 3rd-year undergraduate.

Likelihood is used a lot in phylogeny estimation

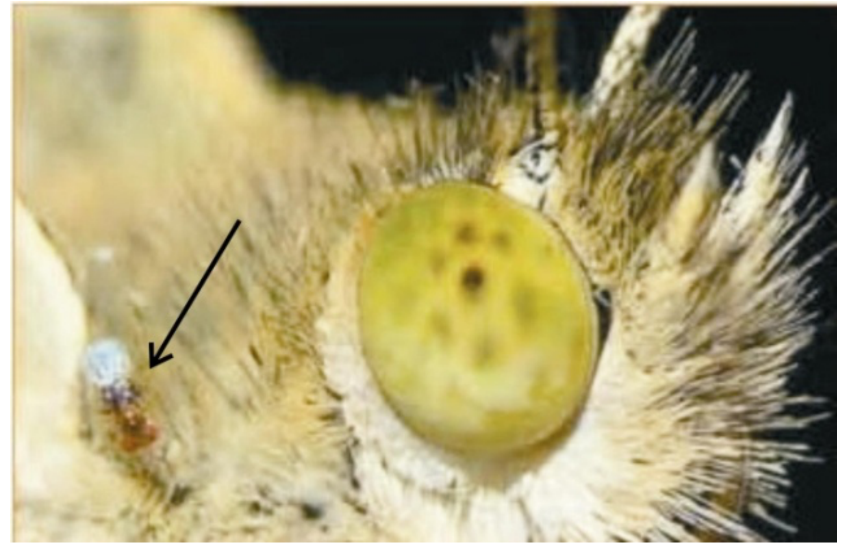
Three proposed trees of ancestor–descendant relationships between humans and the other great apes. The human branch is highlighted. Numbers at the bottom are the likelihoods of each proposed tree based on gene sequence data (Rannala and Yang 1996). The likelihood of the left-most tree is the highest.

L[tree = i | gene sequences] = Pr[gene sequences | tree = i]



What matters is not the likelihood of each tree as such, but the likelihood of each tree <u>relative to the others</u>.

Example 1: Estimate a binomial proportion *p*



Data: The tiny wasp, *Trichogramma brassicae*, rides on female cabbage white butterflies, *Pieris brassicae*. When a butterfly lays her eggs on a cabbage, the wasp climbs down and parasitizes the freshly laid eggs.

Fatouros et al. (2005) carried out trials to determine whether the wasps can distinguish mated female butterflies from unmated females. In each trial a single wasp was presented with two female cabbage white butterflies, one a virgin female, the other recently mated. Y = 23 of 32 wasps tested chose the mated female.

What is the proportion p of wasps in the population choosing the mated female?

Y = 23 “successes”, n = 32 trials. Use these data to estimate p.

Example 1: Estimate a binomial proportion *p*

Likelihood function for the binomial proportion *p*

Data: $Y = 23$, $n = 32$

$$L[p \mid Y \text{ chose mated female}] = \Pr[Y \text{ chose mated female} \mid p]$$

$$L[p \mid 23 \text{ chose mated}] = \binom{32}{23} p^{23}(1-p)^{9}$$

For example, the likelihood of $p = 0.5$, given the data, is

$$L[p = 0.5 \mid 23 \text{ chose mated}] = \binom{32}{23}(0.5)^{23}(1-0.5)^{9} = 0.00653$$

in R:

```
dbinom(23, 32, prob=0.5)
[1] 0.00653062
```

Example 1: Estimate a binomial proportion *p*

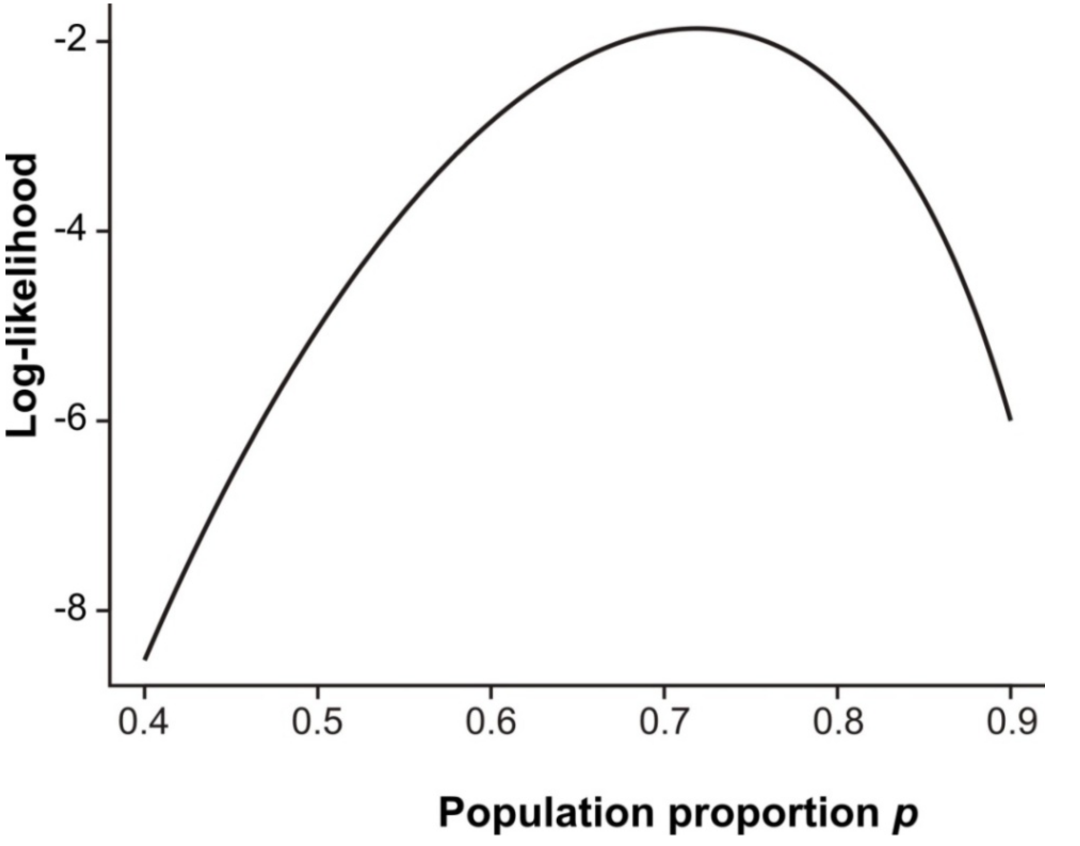
Easier to work with log-likelihoods

$$\ln L[0.5 \mid 23 \text{ chose mated}] = \ln\binom{32}{23} 23 \ln(0.5)\ 9 \ln(1-0.5) = -5.03125$$

in R:

```
dbinom(23, 32, prob = 0.5, log=TRUE)
[1] -5.031253
```

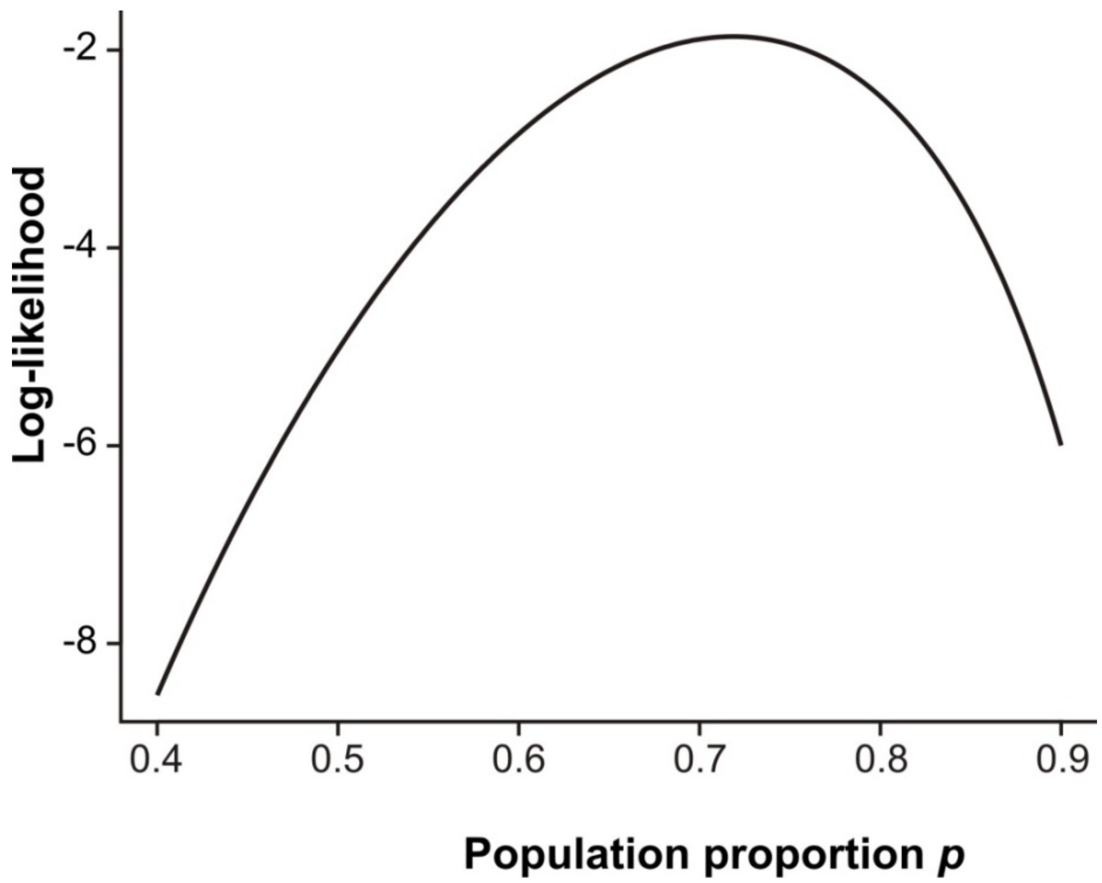
Repeat for many values of *p*
to get the log-likelihood curve:



Likelihood works backward from probability.

We use likelihood to estimate unknown parameters based on *known data*.

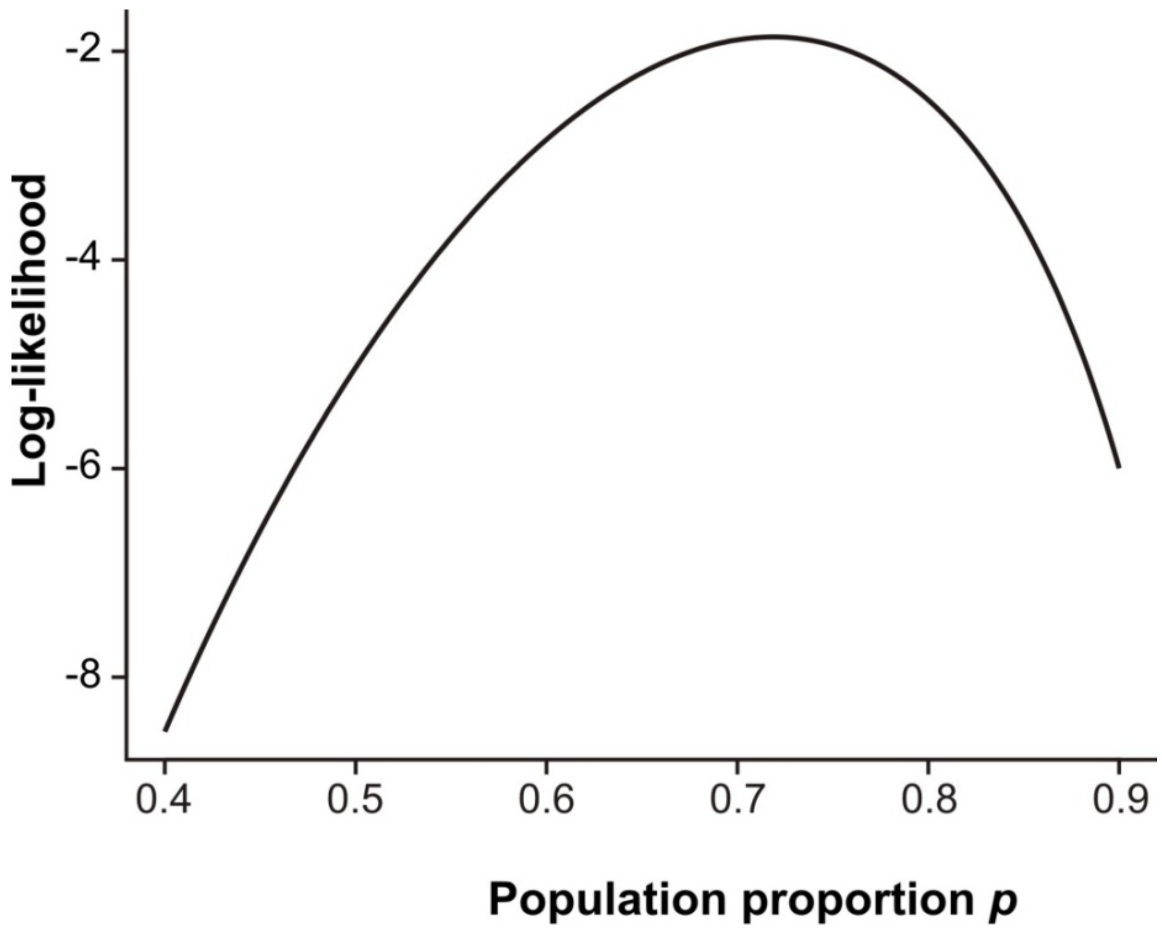
The parameters are treated as variables, the data are a constant, unvarying.



Likelihood works backward from probability.

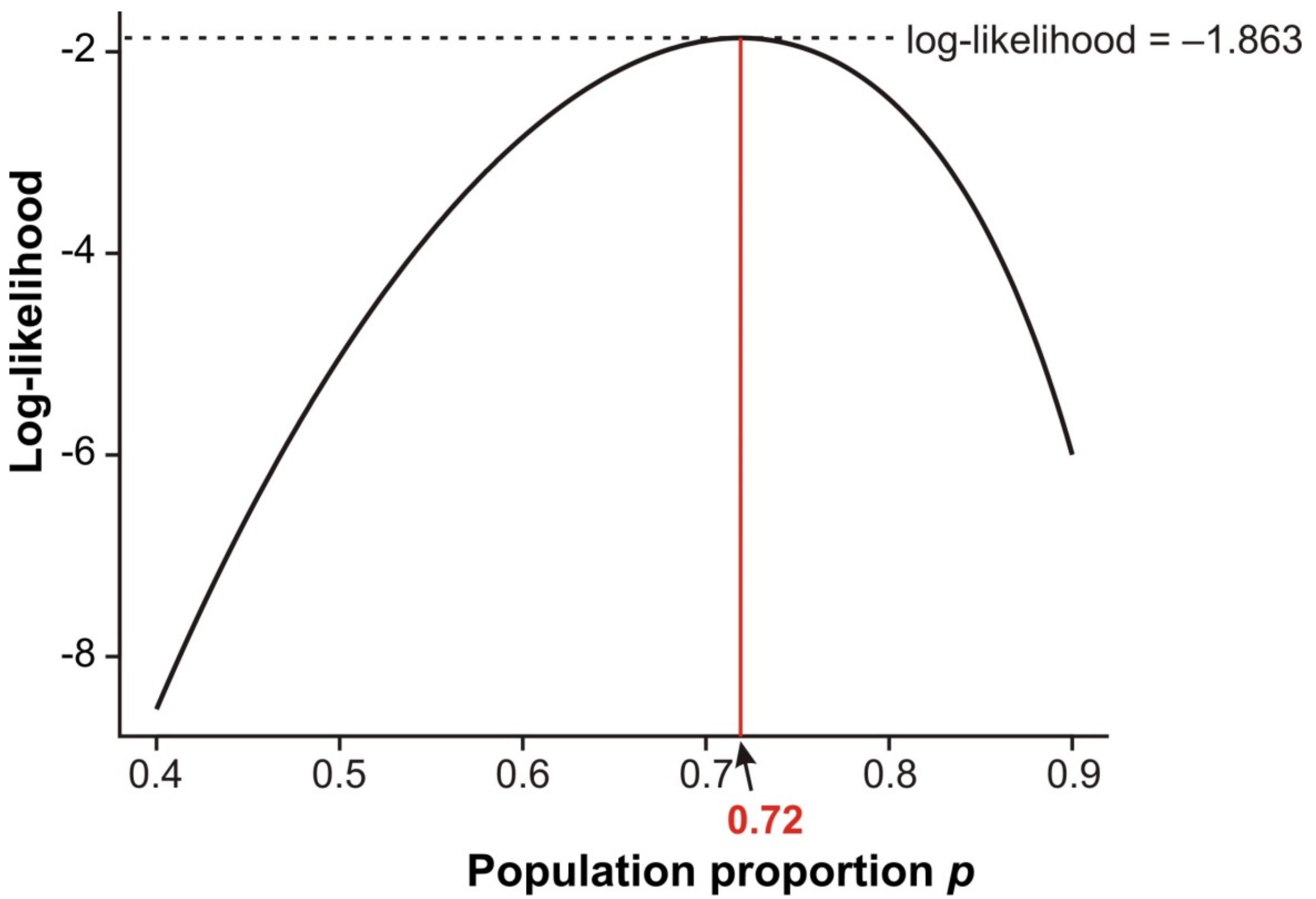
But the likelihood function is not a probability distribution.

The population proportion p is the variable of the function, but it is not a random variable (its value is not determined by random trial).



Maximum likelihood estimate

The likelihood ratio (difference of log-likelihoods) measures relative support for alternative parameter values. The *maximum likelihood estimate* (MLE) of a parameter is the parameter value having the highest likelihood (and log-likelihood), given the data. This is the parameter value most strongly supported by the data.



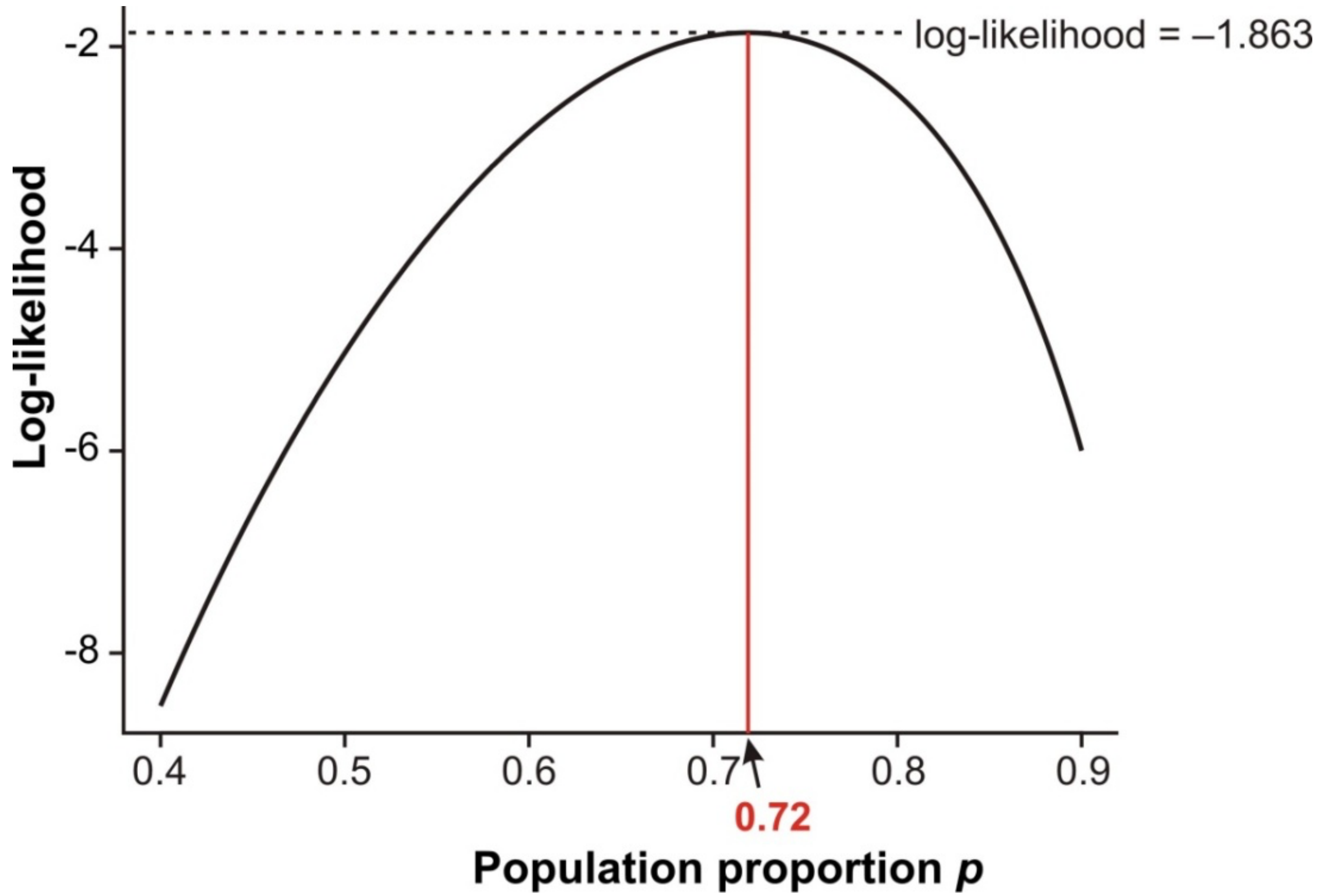
Maximum likelihood estimate

The ML estimate could instead have been obtained more easily as

$$\hat{p} = \frac{Y}{n} = \frac{23}{32} = 0.72$$

The conventional formula for estimating a proportion yields the ML estimate.

Most formulas you use to estimate parameters yield maximum likelihood estimates.



Log-likelihood ratio test

$$G = 2\ln\frac{L[\text{full model} \mid \text{data}]}{L[\text{reduced model} \mid \text{data}]}$$

G is the log-likelihood ratio test statistic.

Under H_0, *G* is approximately χ^2 distributed.

Degrees of freedom for the χ^2 are equal to the difference between the full model and the reduced model in the number of parameters estimated from the data.

Very general method – applies to any data.

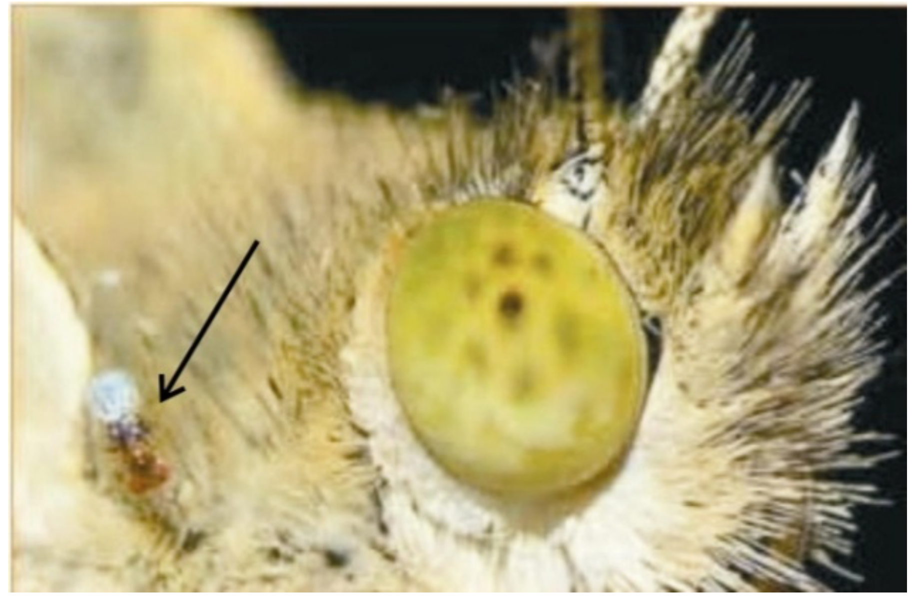
The approximation to the χ^2 distribution improves with increasing sample size.

Log-likelihood ratio test

Example 1: Fatouros et al. (2005) carried out trials to determine whether the wasps can distinguish mated female butterflies from unmated females. In each trial, a single wasp was presented with two female cabbage white butterflies, one a virgin female, the other recently mated. Results: 23 successes out of n = 32 trials.

Reduced model:

H_0: Wasps choose mated and unmated females with equal probability (p = 0.5)



Full model:

H_A: Wasps prefer one type of female over the other ($p \neq 0.5$)

To fit the full model, p is estimated from the data. In this sense, the full model has 1 more term than the reduced model.

Log-likelihood ratio test

$$G = 2 \ln \frac{L[\text{full model} \mid \text{data}]}{L[\text{reduced model} \mid \text{data}]}$$

Applied to the wasp example:

$$G = 2 \ln \frac{L[p = \hat{p} = 0.72 \mid 23 \text{ of } 32 \text{ chose mated female}]}{L[p = p_0 = 0.50 \mid 23 \text{ of } 32 \text{ chose mated female}]}$$

A parameter estimated from the data uses the maximum likelihood estimate (e.g., $\hat{p} = 0.72$ in the full model here).

Log-likelihood ratio test

From calculations using formulae shown earlier,

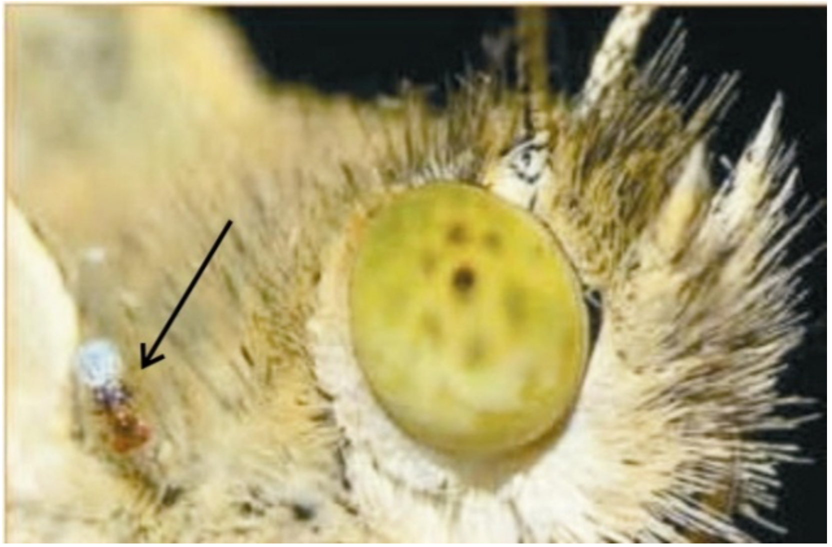
$L[0.72 \mid 23 \text{ of } 32 \text{ chose mated female}] = 0.1553$

$L[0.50 \mid 23 \text{ of } 32 \text{ chose mated female}] = 0.00653$

$$G = 2 \ln \frac{0.1553}{0.00653} = 6.336$$

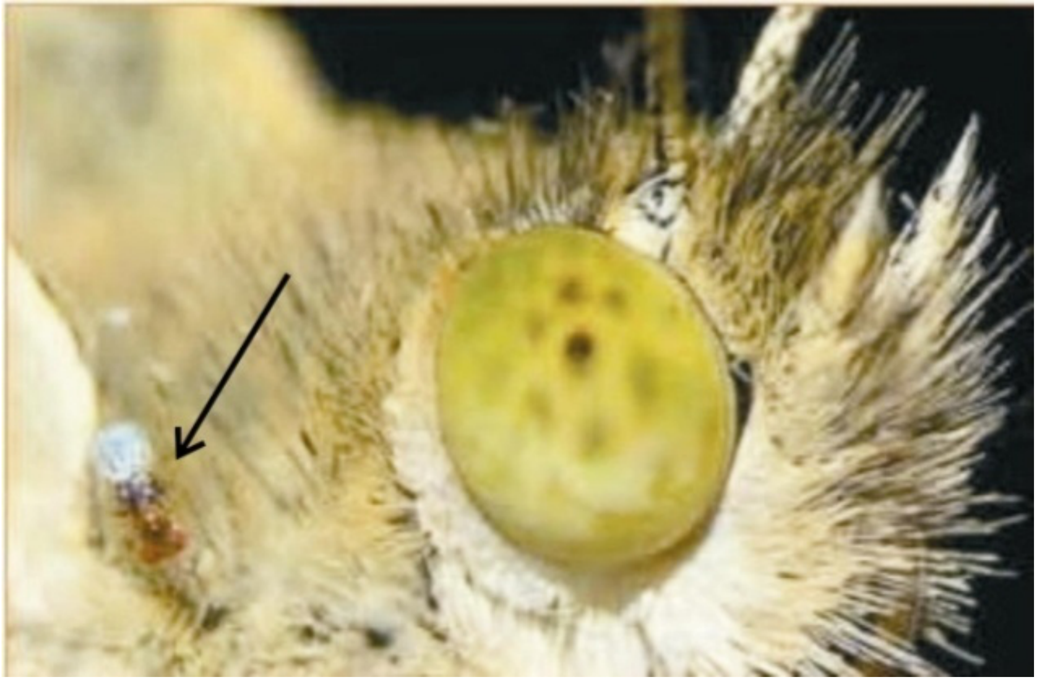
df = 1, so the critical value $\chi^2 = 3.841$

Since 6.336 > 3.841, we reject H_0.

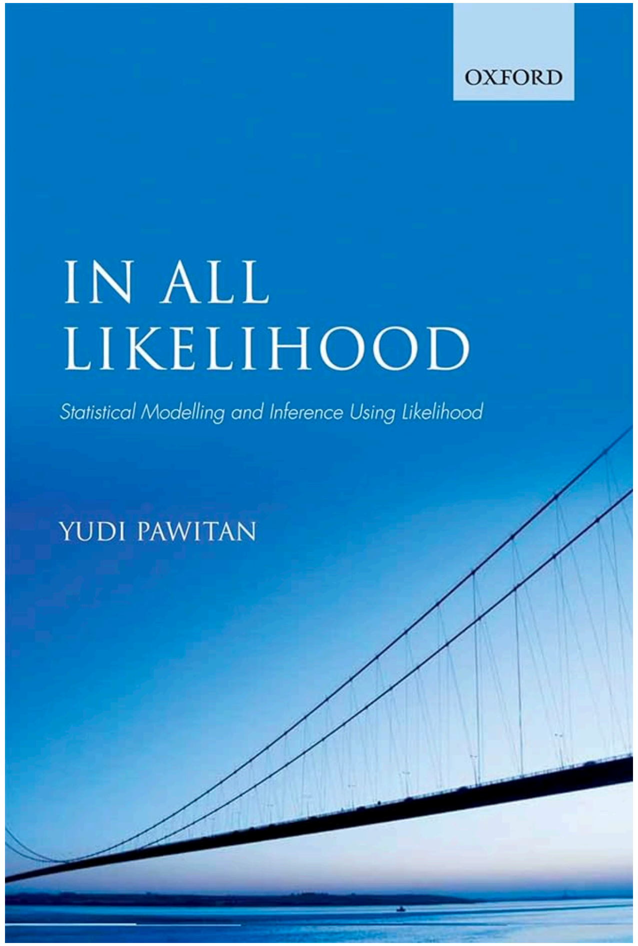


Wasp example: finale

The chemical that the wasps use to distinguish mated from unmated females is benzyl cyanide, which the male butterfly passes to the female during mating. The compound is an “anti-aphrodisiac”, rendering the mated female less attractive to other male butterflies (Fatouros *et al*. 2005).



Readings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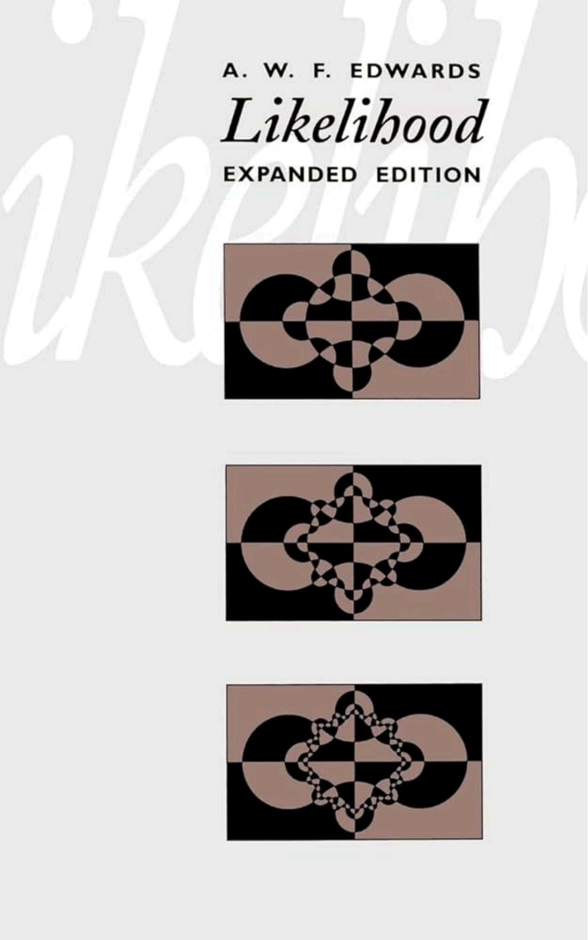
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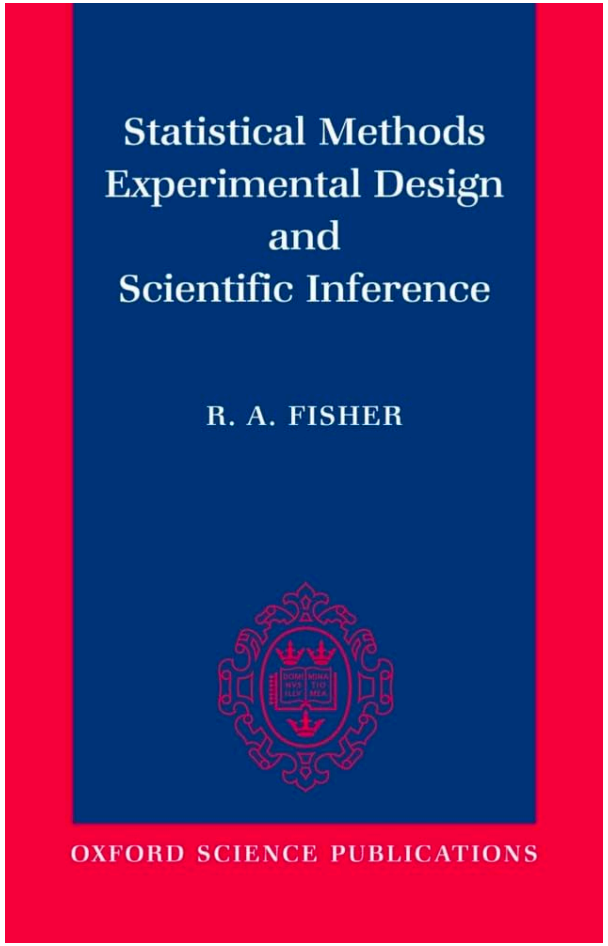
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